

AP Calc Lesson 2.7 Homework

Answer Key

1. $h'(2) = -4(-1) + 2(3) = 10$
2.
 - a. $h(x) = -3x^6$
 - b. $h'(x) = -18x^5$
 - c. $f'(x) = 12x^3$ and $g'(x) = -2x$, so $f'(x) \cdot g'(x) = 12x^3 \cdot -2x = -24x^4$.
 - d. The derivative of a product does not equal the product of the derivatives. The product rule is required when taking the derivative of a product to account for the combined effect of the changes in both functions rather than the individual changes of each function.
3.
 - a. $j(x) = g(x) \cdot x^{-1/2}$
 - b. $j'(x) = g(x) \cdot \left(-\frac{1}{2}\right)x^{-3/2} + x^{-1/2} \cdot g'(x)$
 - c. $j'(1) = 3 \cdot \left(-\frac{1}{2}\right) + 1 \cdot 2 = \frac{1}{2}$
4.
 - a. $\frac{f\left(\frac{3\pi}{2}\right) - f\left(\frac{\pi}{2}\right)}{\frac{3\pi}{2} - \frac{\pi}{2}} = 0$
 - b. $f'(x) = -e^x \sin x + e^x \cos x$
 - c. $x = \frac{5\pi}{4}$
5.
 - a. $g'(x) = -f(x)(6x) + (3x^2 - 5)(-f'(x))$
 - b. $g'(1) = -f(1)(6) + (-2)(-f'(1)) = -12 + 14 = 2$
6.
 - a. $f'(x) = (4 - 3x)e^x + e^x(-3) = -3xe^x + e^x$
 - b. $x = \frac{1}{3}$
 $f'(x) = 0 \Rightarrow -3xe^x + e^x = 0 \Rightarrow e^x(-3x + 1) = 0 \Rightarrow -3x + 1 = 0 \Rightarrow x = \frac{1}{3}$
7.
 - a. $f'(x) = (4x^2 - ax)g'(x) + g(x)(8x - a)$
 - b. $f'(-2) = (16 + 2a)g'(-2) + g(-2)(-16 - a) = (16 + 2a)(3) + (-4)(-16 - a) = 48 + 6a + 64 + 4a = 10a + 112 \Rightarrow 10a + 112 = 2 \Rightarrow 10a = -110 \Rightarrow a = -11$
8.
 - iii. $h'(-1) > 0$
 - b. $h'(x) = f(x)g'(x) + g(x)f'(x) \Rightarrow h'(-1) = f(-1)g'(-1) + g(-1)f'(-1) \Rightarrow (+)(+) + (-)(-) > 0$. Since function f is even, f is positive and decreasing for $x < 0$. The graph of function g can be used to show $g(-1) < 0$ and $g'(-1) > 0$.
9. D
10. C